LECTURE-5

Wednesday
2/9/26.

Tree of Merge Sort:

2^0 $T(n)$ $n \cdot c$

2^1 $T(n/2)$ $T(n/2)$ $(n/2)c + (n/2)c = nc$

2^2 $T(n/4)$ $T(n/4)$ $T(n/4)$ $T(n/4)$ $(n/4)c + (n/4)c + (n/4)c + (n/4)c = nc$

2^{k-1}

$(2)^k$ $T(1)$ . . . . $(T(1))$

↑ these are no. of nodes at each level

⇒ WORK AT NON-Terminal Nodes.

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1) Same work at each level = (Height) * (Work at each level)

2) Diff. work at each level:

→ (i) Work increases level by level = Sum of ↑Geometric progression $\frac{a(r^k-1)}{(r-1)}$ ↳ $r > 1$

(cost of leaf nodes will dominate)

→ (ii) Work decreases level by level = Sum of geometric prog. ↳ $r < 1$ ↳ $\frac{a(1-r^k)}{(1-r)}$

(Cost of root will dominate)

⇒ Work at terminal node

k = height = $\log_b(n)$ (from Binary Search tree) = $\log_2(n)$

No. of terminal/leaf Nodes = 2^k (Came from $2^0, 2^1, 2^2 \ldots 2^k$)

* Plug value of $k = \log_2(n)$

2^k means $\cancel{2}^{\cancel{\log_2}(n)}$ ∴ $2^k = \boxed{n}$ → means 'n' nodes at leaf

WD. at terminal nodes = $\boxed{n \cdot c_1}$ ↗ Constant work at terminal

① $T(n)$ = (WD at terminal) + (WD at non terminal)

$$T(n) = \log_2(n) \cdot nc + nc_1$$

$$T(n) = \Theta(n \log n)$$

Q: $T(n) = \underset{a}{3}\, T(n/\underset{b}{2}) + \Theta(n)$: 3 ↑children nodes ~~at each level~~ & sub problem halves at each level

Formulas:

② WD at terminal = (No. of terminal nodes) * (WD at each terminal Node) ↗ mostly c_1 (Constant)

③ WD at Non terminal = (No. of Non terminal Nodes at each level) * (WD at each node on that non terminal level)

↓ Use arithmetic/geometric Sum formula

Pearl Notes

$\underrightarrow{\text{Imp}}$: $(3)^{\log_2(n)} = (n)^{\log_2(3)}$

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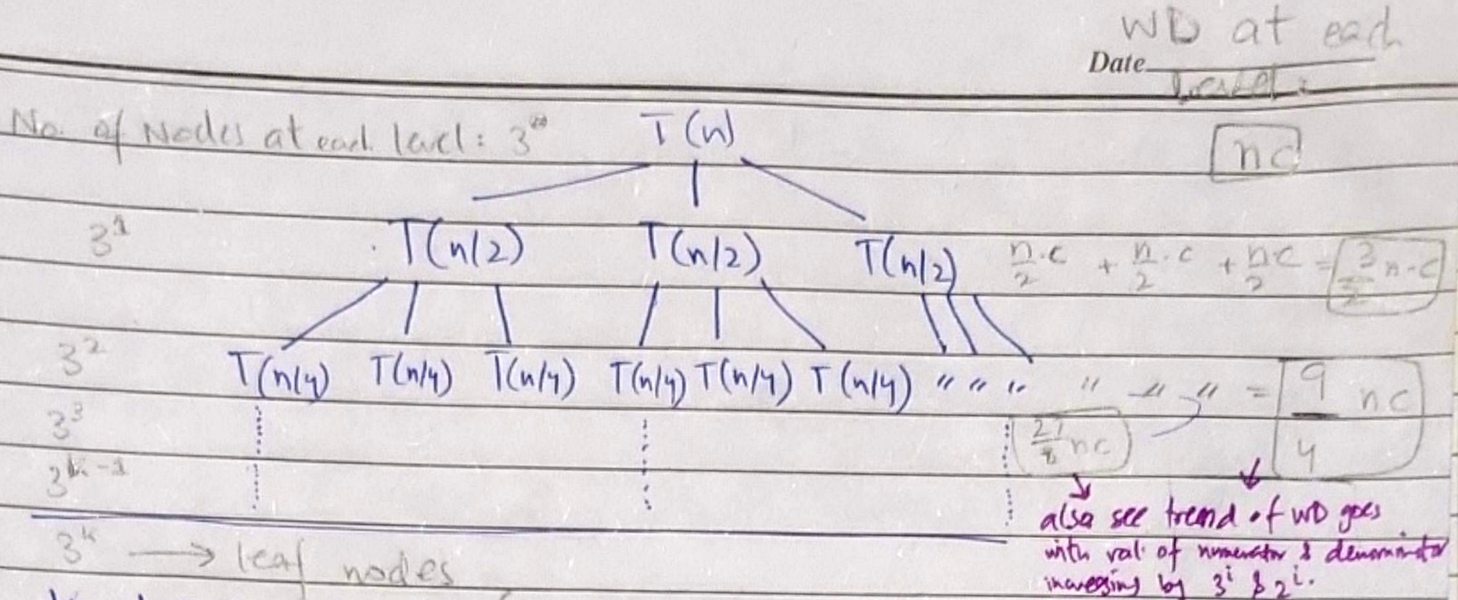


K = height $= \log_b(n)$ (from binary search tree) $= \log_2(n)$

No. of leaf nodes $= 3^k \rightarrow$ ①

★ Put value of 'k' in ① : $3^{\log_2(n)} \xrightarrow{\text{equal to}} \boxed{n^{\log_2(3)}}$

No. of leaf nodes $= n^{\log_2(3)}$

→ Workdone at leaf nodes $= (n)^{\log_2 3} * C1 = \boxed{O\left(n^{\log_2(3)}\right)}$

→ Workdone at non-terminal nodes = (no. of Nodes at each level) * (WD at each node on that level)

$= \left(3^i * \frac{n}{2^i}\right)$

★ Sum of all WD at all non terminal level is found by sum of geometric series as WD at each level is : n/c, $\frac{3}{2}nc$, $\frac{9}{4}nc$, $\frac{27}{8}nc$ ∴

$r = \frac{9}{4}nc \div \frac{3}{2}nc = \boxed{3/2}$

Sum of geometric $= \frac{a(r^n - 1)}{r - 1} = \frac{nc\left[(3/2)^k - 1\right]}{(3/2 - 1)}$ height of tree $\rightarrow k = \log_2 n$

height of tree

$= \frac{nc\left[(3/2)^{\log_2(n)} - 1\right]}{3/2 - 1}$

=

* denominator reciprocal & brought '2'

$$= 2nc\left(\frac{3^{\log_2 n}}{\cancel{2^{\log_2 n}}} - 1\right)$$

$$= 2nc\left(\frac{3^{\log_2 n}}{n} - 1\right) \longrightarrow 2\cancel{n}c \cdot \frac{3^{\log_2 n}}{\cancel{n}} - 2nc$$

$$\boxed{= 2c \cdot 3^{\log_2 n} - 2nc}$$

$$= 2c \cdot (n)^{(\log_2(3))} - 2nc$$

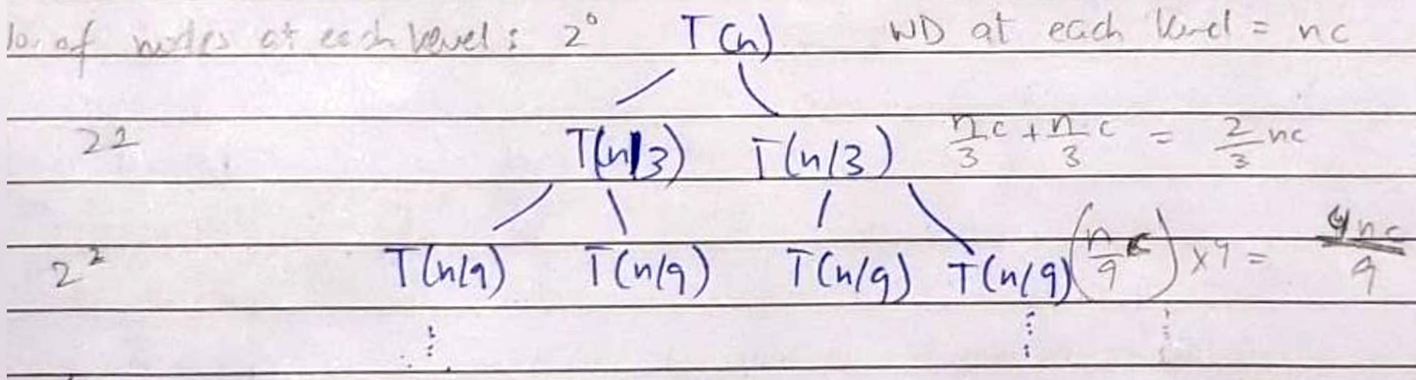
$\therefore$ WD. at Non terminal Nodes $= \Theta\left((n)^{\log_2 3}\right)$

$\Rightarrow T(n) =$ WD at terminal + WD at Non-terminal

$$= \Theta(n)^{\log_2(3)} + \Theta(n)^{\log_2(3)}$$ } Same Θ values

$$\boxed{T(n) = \Theta(n)^{\log_2(3)}}$$ answer.

Q: $T(n) = \underset{a}{2}\,T(n/\underset{b}{3})$: 2 children nodes & sub problems get one-third



2^k = leaf nodes

WD at terminal is calculated the same way like $2^{\log_3 n} \rightarrow \boxed{n^{\log_3(2)} * C_1}$

WD at Non terminal :- $\frac{4}{9}nc \div \frac{2}{3}nc \longrightarrow \boxed{r = \frac{2}{3}}$

$$= a\left(\frac{1-r^n}{1-r}\right) \longrightarrow n = k = \log_3(n)$$

$$= nc.\frac{\left[1 - (2/3)^{\log_3(n)}\right]}{[1 - 2/3]}$$

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$$= nc \cdot \left[\frac{1 - \frac{(2)^{\log_3(n)}}{3^{\log_3(n)}}}{1/3} \right] = 3nc \cdot \left(1 - \frac{2^{\log_3(n)}}{\cancel{3}^{\cancel{\log_3}(n)}} \right)$$

$$= 3nc \left(1 - \frac{n^{\log_3(2)}}{n} \right)$$

$$= 3nc - 3\cancel{n}c \cdot \frac{n^{\log_3(2)}}{\cancel{n}} = \boxed{3nc - n^{\log_3(2)}}$$

$$T(n) = WD_{terminal} + WD_{non\ terminal}$$

$$= \left[n^{\log_3(2)} * c_1 \right] + \left[3nc - n^{\log_3(2)} \right]$$

* in terms of Θ

$$\boxed{T(n) = \Theta\left(n^{\log_3(2)}\right)}$$